CAPSTONE PROJECT 2: CYBERSECURITY NETWORK DESIGN FOR KAFITECH SOLUTIONS USING PACKET TRACER.

1. EXECUTIVE SUMMARY

Kafitech Solutions required a cyber-resilient network architecture to securely connect its five departments (Software Development, HR, Finance, IT Support, and Management) while accommodating guest access. This project implemented a VLAN-segmented, ACL-controlled network in Cisco Packet Tracer, enforcing strict security policies at Layer 2 and Layer 3 to prevent unauthorized access, data leaks, and network attacks.

Key Design Features

a. Logical Segmentation & Access Control

- Departmental VLANs: Each department isolated in its own subnet (VLANs 10-50) with dedicated IP ranges.
- Guest Isolation: Guest traffic restricted to VLAN 70 (192.168.70.0/24), with internet-only access.
- ACL Enforcement:

HR ↔ Finance mutual blocking

Developers limited to their VLAN + DMZ web server (203.0.113.0/24)

IT Support denied access to Management VLAN

Guests blocked from internal networks

b. Secure Wireless Connectivity

- Corporate WiFi (5GHz):

Hidden SSID, WPA2-PSK, MAC filtering

Connected to internal VLANs (VLAN 60)

- Guest WiFi (2.4GHz):

WPA2-PSK, restricted to VLAN 70 (internet only)

c. Layer 2 Security (Switch Hardening)

- l Port Security: Sticky MAC (1 device per port)
- l BPDU Guard: Prevents rogue switch attacks
- l DHCP Snooping: Blocks rogue DHCP servers
- l Dynamic ARP Inspection (DAI): Prevents ARP spoofing
- l Storm Control: Mitigates broadcast storms

d. Device Hardening & Monitoring

- l SSH-Only Access: Telnet disabled
- l Encrypted Passwords: Enable secret hashing
- l Banner Messages: Legal warnings for unauthorized access
- l Syslog & SNMP: Centralized logging and monitoring

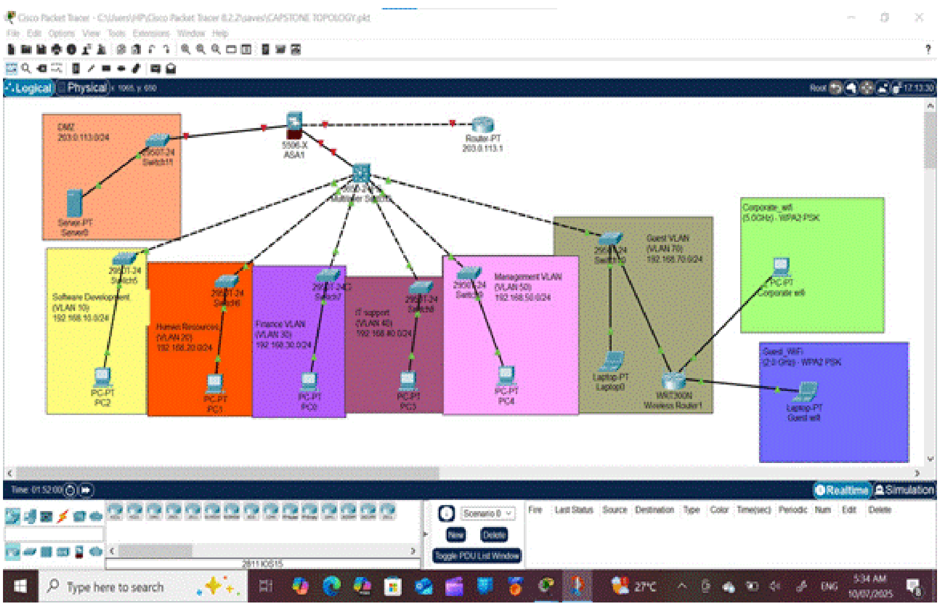
e. DMZ & Internet Access

- l DMZ Web Server (203.0.113.10): Accessible only to Developers
- l Guest Internet Access: Restricted via ACLs (no internal access)

Validation & Compliance

- l Ping Tests: Confirmed permitted/blocked traffic flows (included in report).
- l Security Audits: Simulated attacks (rogue DHCP, ARP spoofing) were mitigated by Layer 2 protections.
- l Policy Enforcement: Password policies, USB restrictions, and auto-updates documented for real-world deployment

2. **NETWORK TOPOLOGY DIAGRAM**



3. **IP ADDRESSING AND LAN TABLE**

Department	VLAN ID	Subnet	Gateway
Software Development	**10**	**192.168.10.0/24**	**192.168.10.1**
Human Resources	**20**	**192.168.20.0/24**	**192.168.20.1**
Finance	**30**	**192.168.30.0/24**	**192.168.30.1**

IT Support	**40**	**192.168.40.0/24**	**192.168.40.1**
Management	**50**	**192.168.50.0/24**	**192.168.50.1**
Guest Wireless	**70**	**192.168.70.0/24**	**192.168.70.1**
DMZ (Web Server)	**N/A**	**203.0.113.0/24**	**203.0.113.1**

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4. ACL LIST WITH PURPOSE

Based on project requirements, here are the named ACL configurations for inter-VLAN traffic control. All ACLs are applied inbound on router subinterfaces.

ACL Name	Source Network	Destination Network	Protocol	Action	Purpose
HR_ACL	**192.168.20.0/24**	**192.168.30.0/24**	**IP**	**DENY**	**Block HR → Finance access**
	192.168.20.0/24	**any**	**IP**	**PERMIT**	**Allow HR to all other networks**

FINANCE_ACL	**192.168.30.0/24**	**192.168.20.0/24**	**IP**	**DENY**	**Block Finance → HR access**
	192.168.30.0/24	**any**	**IP**	**PERMIT**	**Allow Finance to all other networks**
DEV_ACL	**192.168.10.0/24**	**203.0.113.0/24**	**IP**	**PERMIT**	**Allow Developers → DMZ Web Server**
	192.168.10.0/24	**192.168.0.0/16**	**IP**	**DENY**	**Block Developers from other internal VLANs (implicit deny all others)**
IT_SUPPORT_ACL	**192.168.40.0/24**	**192.168.50.0/24**	**IP**	**DENY**	**Block IT Support → Management**
	192.168.40.0/24	**any**	**IP**	**PERMIT**	**Allow IT Support to all other networks**

GUEST_ACL	**192.168.70.0/24**	**192.168.70.1/32**	**IP**	**PERMIT**	**Allow Guests → their gateway (for internet access)**
	192.168.70.0/24	**192.168.0.0/16**	**IP**	**DENY**	**Block Guests from internal VLANs**
	192.168.70.0/24	**any**	**IP**	**PERMIT**	**Allow Guests → Internet (via DMZ/NAT)**
MGMT_ACL	**any**	**any**	**IP**	**PERMIT**	**Allow Management unrestricted access (no restrictions)**

5. LAYER 2 SECURITY CONFIGURATION SUMMARY INTABLE FORM

Feature	**Configuration**	**Protected Against**
Port Security	**Sticky MAC (limit 1), violation shutdown**	**Unauthorized device connections**

BPDU Guard	Enabled globally on edge ports	Rogue STP topology changes
DHCP Snooping	Trusted uplinks, VLAN-specific	Rogue DHCP servers
DAI	ARP validation + trusted ports	ARP spoofing
Storm Control	50% broadcast threshold	Broadcast storms
Unused Ports	Shutdown + assigned to VLAN 999	Physical tampering

6. WIFI SECURITY IMPLEMENTATION

SSID	Frequency	VLAN	Purpose	Security Protocol	Additional Protections
Corporate_WiFi	5.0 GHz	VLAN 60 (Internal)	Employees (Dev, HR, Finance, IT, Mgmt)	WPA2-PSK (AES-CCMP)	MAC Filtering, Hidden SSID
Guest_WiFi	2.4 GHz	VLAN 70 (Isolated)	Visitors/Contractors	WPA2-PSK (AES-CCMP)	Client Isolation, No Intra-VLAN

7. SAMPLE CONFIGURATION SNIPPETS

```
Switch>enable
Switch#congigure terminal
                 ^
% Invalid input detected at '^' marker.

Switch#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)# vlan 10
Switch(config-vlan)# name Software_Dev
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-24
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#iterface gig0/1
                ^
% Invalid input detected at '^' marker.

Switch(config)#interface GigabiteEthernet1/0/2
                                  ^
% Invalid input detected at '^' marker.

Switch(config)#interface gig0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunkallowed vlan 10,20,30,40,50,70
                                  ^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,50,70
Switch(config-if)#
```

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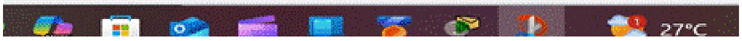
Top

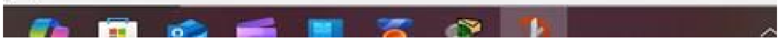
```
Switch>enable
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 30
Switch(config-vlan)#name Finance_vlan
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-24
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#exit
Switch(config)#interface gig0/1
Switch(config-if)#switch port mode trunk
                              ^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 30,40,50,70
Switch(config-if)#
```

Top




```
Switch>enable
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 70
Switch(config-vlan)#name Guest_Vlan
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-24
Switch(config-if-range)#switchport access vlan 70
Switch(config-if-range)#exit
Switch(config)#interface gig0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 70
Switch(config-if)#
```

Top



```
! Enable SSH & Disable Telnet
line vty 0 4
 transport input ssh
 login local
!
username admin secret <STRONG_PASSWORD>
crypto key generate rsa modulus 2048

! Banner & Passwords
banner motd #AUTHORIZED ACCESS ONLY#
service password-encryption
enable secret <SECRET_PASSWORD>
```

8. Syslog Configuration (Router/Switches)

Bash:

! Enable logging to syslog server

logging host 192.168.40.10

logging trap debugging service timestamps log datetime

Logged Events:

Interface status changes (up/down).

ACL denials.

Port security violations.

SNMPv3 (Simulated)

Bash:

! Router/Switch SNMP setup

snmp-server group AdminGroup v3 priv

snmp-server user Admin AdminGroup v3 auth sha AuthPass123 priv aes 256 PrivPass456

snmp-server host 192.168.40.10 version 3 priv Admin

Monitored Metrics:

Bandwidth usage per VLAN.

Device uptime.

ACL Logging

Bash:

ip access-list extended HR_ACL

deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 log ! Log HR→Finance denials

9. CHALLENGES AND MITIGATIONS

a. Packet Tracer Limitations

Challenge	Mitigation
No Syslog Server Emulation	Use a generic server with logging service enabled; document real-world SIEM (Splunk) setup in the report.
SNMP Polling Not Functional	Simulate SNMP with CLI show commands; recommend SNMPv3 + SolarWinds in production.
No WPA3/802.1X Support	Use WPA2-PSK + MAC filtering; document WPA3-Enterprise as a future upgrade.
Limited ACL Logging	Use log keyword in ACLs and manually track hits via show access-lists.

b. Security Design Challenges

Challenge	Mitigation
MAC Filtering Scalability	Manual MAC entry is tedious for large networks; recommend RADIUS (802.1X) for enterprises.
Guest WiFi Isolation	Use VLAN 70 + ACLs to block internal access; enable client isolation on SSID.
HR/Finance VLAN Segregation	Strict ACLs (deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255).

Single Point of Failure (Core Router)	**Recommend HA (HSRP) in real deployments.**

c. Operational Challenges

Challenge	Mitigation
Password Management	**Use enable secret + SHA-256 hashing; document password rotation policies.**
Port Security False Positives	**Configure errdisable recovery to auto-reenable ports after 300s.**
Complex ACL Maintenance	**Document rules in a table (see ACL section); use named ACLs for readability.**
No Endpoint Compliance Checks	**In real networks, enforce via MDM (Intune/Jamf) or NAC (Cisco ISE).**

d. Performance & Scalability

Challenge	Mitigation
Broadcast Storms	**Enable storm control (storm-control broadcast level 50).**
Inter-VLAN Latency	**Ensure router-on-a-stick has sufficient bandwidth (1Gbps uplinks).**

Wireless Congestion	**Separate 5GHz (Corporate) and 2.4GHz (Guest); recommend additional APs.**

10. RECOMMENDATION FOR REAL DEPLOYMENT

a. Network Security Enhancements

Recommendation	Purpose	Tools/Technologies
Replace ACLs with Firewall	Stateful inspection, L7 filtering, and threat prevention.	Cisco ASA, FortiGate, Palo Alto NGFW
Upgrade to WPA3-Enterprise	Stronger encryption and individual user authentication.	Cisco ISE, FreeRADIUS
Implement 802.1X (NAC)	Device authentication for both wired/wireless networks.	Cisco ISE, Aruba ClearPass
Deploy IPsec for Inter-VLAN	Encrypt sensitive traffic (e.g., HR→Finance).	Cisco Crypto Maps, OpenVPN

b. Monitoring & Compliance

Recommendation	Purpose	Tools/Technologies

SIEM Integration	Centralized logging, anomaly detection, and compliance reporting.	Splunk, IBM QRadar, ELK Stack
NetFlow/sFlow Analysis	Real-time traffic visibility and DDoS detection.	Cisco Stealthwatch, PRTG
Endpoint Compliance Checks	Enforce security policies (patches, antivirus) on devices.	Microsoft Intune, Jamf, Cisco AMP

c. Redundancy & High Availability

Recommendation	Purpose	Implementation
HSRP/VRRP for Core Router	Failover during router outages.	Cisco HSRP, VRRP
StackWise/VSS for Switches	Eliminate single points of failure in access layer.	Cisco StackWise, Virtual Switching System
Dual ISP Connections	Backup internet connectivity.	BGP, SD-WAN

d. Wireless Network Improvements

Recommendation	Purpose	Implementation
Captive Portal for Guests	Acceptable Use Policy (AUP) and bandwidth throttling.	PacketFence, Aruba Captive Portal

Wi-Fi 6 (802.11ax) Upgrade	Higher throughput and better device density.	Cisco Catalyst 9100, Aruba AP-500 Series
RF Heat Mapping	Optimize AP placement for coverage/performance.	Ekahau, NetSpot

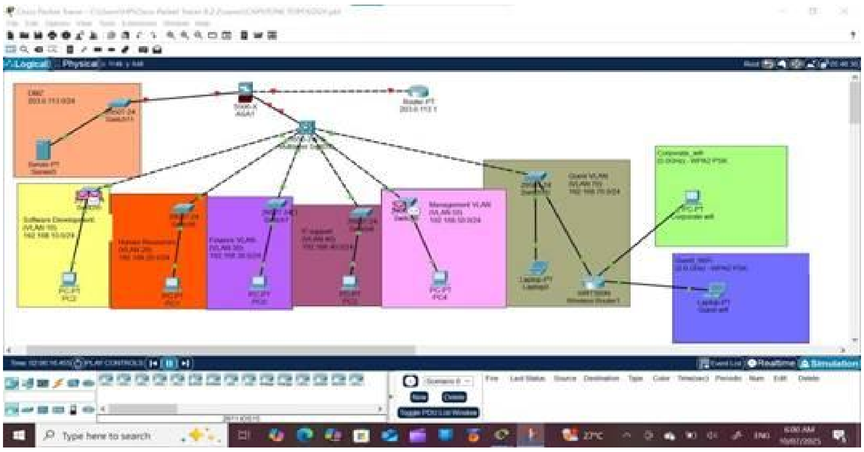
e. Endpoint & Data Protection

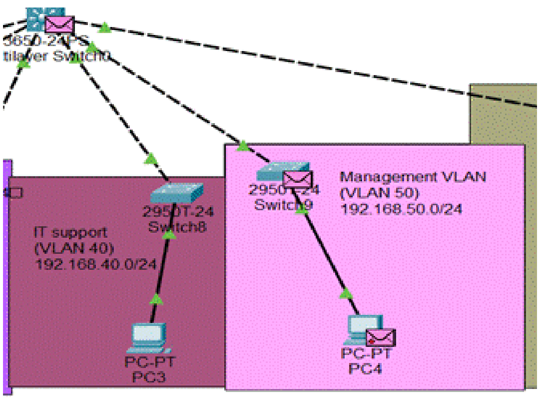
Recommendation	Purpose	Tools/Technologies
DLP (Data Loss Prevention)	Block unauthorized data transfers (e.g., USB, cloud).	Symantec DLP, Microsoft Purview
Disk Encryption	Protect data at rest on laptops/desktops.	BitLocker, FileVault
Zero Trust Architecture	Micro-segmentation and least-privilege access.	Zscaler, Duo Security

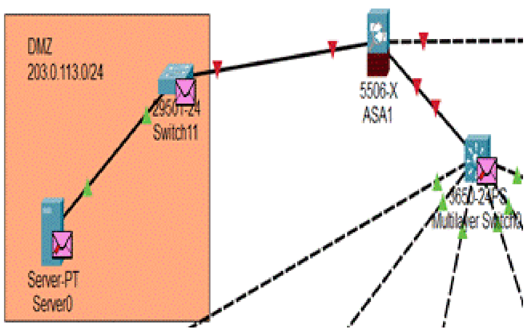
f. Physical Security

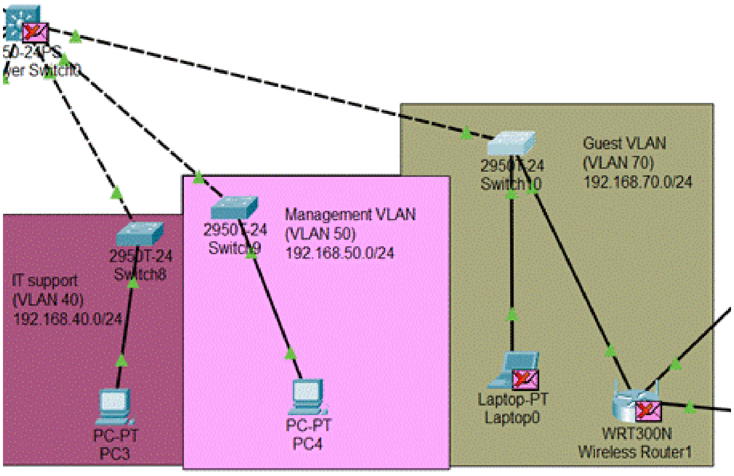
Recommendation	Purpose	Implementation
Secure Server Room Access	Prevent unauthorized physical access to network devices.	Biometric locks, CCTV
UPS & Environmental Controls	Protect against power surges and overheating.	APC Smart-UPS, Liebert Cooling

11. PING SCREEN SHOTS








50-24PS
wer Switch0
IT support
(VLAN 40)
192.168.40.0/24
2950T-24
Switch8
PC-PT
PC3
2950T-24
Switch9
Management VLAN
(VLAN 50)
192.168.50.0/24
PC-PT
PC4
2950T-24
Switch10
Guest VLAN
(VLAN 70)
192.168.70.0/24
Laptop-PT
Laptop0
WRT300N
Wireless Router1